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AICO: A Hierarchical Bayesian Observatory for Enterprise AI Capability Measurement

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Version 1.0 — reproducibility manifests for every experimental table are generated by src/aico/experiments/ and shipped alongside the artifacts.

Abstract

Enterprises evaluate AI models against ever-changing portfolios of public and internal benchmarks, then make procurement, deployment, and research-investment decisions from those evaluations. The dominant methodology — averaging benchmark scores — conflates benchmark difficulty with model ability, ignores measurement error, breaks whenever a benchmark is added or retired, and provides no uncertainty for the decisions it feeds. We present AICO, an open-source scientific platform that treats capability measurement as a latent-variable inference problem. AICO combines (i) a four-layer hierarchical Bayesian model whose benchmark edge is an item response theory (IRT) link, unifying five observation families (normal, beta, binomial, Bernoulli, ordered-logistic) under a single generalized linear latent-trait model; (ii) a scale-stability engine that equates capability scales across benchmark portfolio changes using anchor models; (iii) state-space and Gaussian-process capability forecasting with changepoint detection; (iv) Shapley-based capability attribution; (v) benchmark portfolio optimization; and (vi) graph- and decision-intelligence layers that turn posterior capability estimates into ranked, uncertainty-aware recommendations. We prove identifiability of the hierarchy under four explicit constraints, validate the system with parameter-recovery, posterior-calibration, coverage, and posterior-predictive studies on synthetic ecosystems drawn from the model's own prior at three scales (10^2, 10^3, 10^4 models and benchmarks), ablate each major component, and compare against eight baselines (mean, weighted mean, median, Elo, TrueSkill, Bradley–Terry, single-factor analysis, and classical/multidimensional IRT variants) on ranking consistency, calibration, robustness to missing data, and stability under benchmark addition and retirement. All experiments are reproducible from committed manifests recording code revision, dependency versions, seeds, and hardware; scientific quality is enforced continuously in CI, which fails the build if parameter recovery or calibration degrades.

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1. Introduction

Modern AI development runs on benchmarks. A single enterprise may track hundreds of them: public leaderboards, purchased evaluation suites, and internal workload-specific tests. Each produces numbers on its own scale, with its own noise level, difficulty, and failure modes; models are observed on overlapping but incomplete subsets; and the portfolio itself churns as benchmarks saturate, leak, or are replaced. The operative question — which model is most capable for our purposes, with what confidence, and how is that changing over time? — is a statistical inference problem. Yet the standard practice is to average whatever scores happen to be available.

Averaging fails in specific, consequential ways:

1. Difficulty confounding. A model evaluated mostly on hard benchmarks looks worse than one evaluated on easy benchmarks, regardless of ability.

2. No measurement model. A 500-item benchmark and a 20-item benchmark count equally; a low-discrimination benchmark (one that barely separates models) counts as much as a sharp one.
3. Portfolio fragility. Adding or retiring a benchmark moves every model's average, so scores are not comparable over time — precisely when trend information matters most.
4. No uncertainty. Downstream decisions (procurement, deployment gates, research bets) receive point estimates with no notion of confidence, making risk-aware decision rules impossible.

AICO addresses these failures by modeling the data-generating process explicitly. Its core is a four-layer hierarchical Bayesian model — global capability → source (public / internal) capability → domain capability → benchmark capability — whose benchmark edge is a 1PL/2PL/3PL IRT link with per-benchmark discrimination, difficulty, and (optionally) guessing parameters, and whose observation layer accommodates five likelihood families simultaneously. One posterior fit yields every reported quantity: global and per-layer capability indices, a public-vs-internal generalization gap, benchmark difficulty and discrimination estimates, full posterior uncertainty, and calibrated ranks.

Around this core, AICO builds the machinery an observatory needs and a single model does not: scale stability across portfolio churn, forecasting of capability trajectories, attribution of capability changes, optimization of the benchmark portfolio itself, a capability intelligence graph, and an executive decision layer. Each is described in Section 5 and ablated in Section 8.

Contributions.

- A unified generalized linear latent-trait formulation (Section 4.4) covering mixed metric types per benchmark, with an explicit identifiability proof (Appendix A.3).
- An open, tested implementation: ~34k lines of typed Python (PyMC/NUTS and ADVI inference), 95%+ line-coverage CI, strict static typing.
- A validation methodology run continuously in CI: parameter recovery, posterior calibration (PIT), HDI coverage, posterior predictive checks, rank stability, and drift detection against a committed metrics baseline (Section 7).
- A simulation framework generating self-consistent synthetic ecosystems from the model's own prior at 10^2–10^4 scale with exact ground truth (Section 6).
- A baseline comparison harness and ablation framework producing the publication tables in Sections 8–9, each shipped with a reproducibility manifest.

2. Motivation

Three audiences drive the requirements.

Enterprise engineering needs one defensible number per model per domain — stable across benchmark churn, honest about uncertainty, and cheap to update when new evaluations arrive. Section 5.2's scale-stability engine and the streaming observatory service address this.

Research needs the measurement layer to be correct: identifiable, calibrated, and validated, so that findings about model capability trends are findings about models and not artifacts of the scoring pipeline. Sections 6–9 exist for this audience.

Executives need decisions, not posteriors: which model to procure, when a capability threshold will be crossed, which benchmarks are worth their evaluation budget. The decision-intelligence layer (Section 5.7) consumes posterior capability distributions and produces ranked recommendations with explicit risk terms; the paper evaluates it by selection regret (Section 8).

3. Related Work

3.1 Psychometrics, IRT, and educational testing

Item response theory (Lord & Novick, 1968; Rasch, 1960; Birnbaum, 1968) models the probability of a correct response as a function of latent ability and item parameters — exactly the confounds (1)–(2) above. The 1PL/2PL/3PL family is standard in educational testing (van der Linden & Hambleton, 1997); test equating and scale linking (Kolen & Brennan, 2014) is the psychometric ancestor of AICO's scale-stability engine, and anchor- item designs correspond to AICO's anchor models. Bayesian IRT with MCMC dates to Albert (1992) and Patz & Junker (1999); the identification-via-prior argument AICO uses is the one articulated by Bafumi, Gelman, Park & Kaplan (2005). Multidimensional IRT (Reckase, 2009) generalizes ability to a vector; AICO instead structures multidimensionality hierarchically (source → domain), which preserves interpretability and gives partial pooling across the hierarchy — a modeling choice we compare empirically in Section 8.

3.2 Latent variable models and Bayesian hierarchies

The metric layer is confirmatory factor analysis (Jöreskog, 1969) with the benchmark capability as the common factor; Rule 4 of the identifiability argument is the classical single-indicator constraint from structural equation modeling (Bollen, 1989). Hierarchical Bayesian modeling and its pooling behavior follow Gelman & Hill (2007); the LKJ prior on correlated layer residuals is Lewandowski, Kurowicka & Joe (2009); non-centered parameterization follows Betancourt & Girolami (2015). Inference uses NUTS (Hoffman & Gelman, 2014) in PyMC (Abril-Pla et al., 2023) with ADVI (Kucukelbir et al., 2017) as the fast path; model criticism uses WAIC (Watanabe, 2010) and PSIS-LOO (Vehtari, Gelman & Gabry, 2017).

3.3 Rating systems

Elo (1978) performs stochastic-gradient-like updates on pairwise outcomes; it is order- dependent, has no principled uncertainty, and discards magnitude information. Glickman (1999) and TrueSkill (Herbrich, Minka & Graepel, 2007) add Gaussian uncertainty via Bayesian filtering, and TrueSkill 2 (Minka et al., 2018) extends the observation model. Bradley–Terry (1952; Hunter, 2004) is the static ML analogue. All three families reduce rich scored observations to pairwise comparisons and model a single scalar skill with no benchmark structure — no difficulty, no discrimination, no domain decomposition. They are included as baselines in Section 8; LLM-arena leaderboards built on them (Chiang et al., 2024) inherit these limitations.

3.4 AI benchmarking and evaluation science

The instability and rapid saturation of AI benchmarks is documented by Ott et al. (2022) and Martínez-Plumed et al. (2021); benchmark contamination and leakage by Zhou et al. (2023). IRT has been applied to NLP evaluation by Lalor et al. (2016) and to LLM leaderboards by Rodriguez et al. (2021) and Polo et al. (2024, tinyBenchmarks), demonstrating that item-level modeling recovers more information from fewer evaluations — consistent with our statistical-efficiency results (Section 8). HELM (Liang et al., 2022) and the BIG-bench analysis (Srivastava et al., 2022) motivate multi-metric, multi-domain measurement; AICO differs in fitting a single generative model across the portfolio rather than reporting per-scenario tables. Benchmark governance — versioning, deprecation, audit trails — follows the direction of datasheets and model cards (Gebru et al., 2021; Mitchell et al., 2019), implemented in AICO's governance module.

3.5 Forecasting and decision intelligence

Capability forecasting uses classical structural time series (Harvey, 1989; local linear trend Kalman filtering), GP regression (Rasmussen & Williams, 2006), HMM regime detection (Rabiner, 1989), and logistic/power-law frontier fits in the spirit of performance-curve extrapolation (Hestness et al., 2017; Kaplan et al., 2020). Forecast evaluation follows Gneiting & Raftery (2007): proper scoring rules (CRPS) and interval coverage. The decision layer draws on multi-criteria decision analysis (Keeney & Raiffa, 1976) and Bayesian decision theory (Berger, 1985); attribution uses Shapley values (Shapley, 1953; Lundberg & Lee, 2017 for the modern estimation toolkit).

3.6 Comparison summary

Method	Difficulty modeled	Uncertainty	Missing data	Portfolio-change stability	Multi-metric	Hierarchical structure
Simple / weighted average	✗	✗	biased	✗	ad hoc	✗
Median	✗	✗	biased	✗	ad hoc	✗
Elo	implicit, discarded	✗	order-sensitive	✗	✗	✗
TrueSkill	implicit, discarded	filtered Gaussian	order-sensitive	✗	✗	✗
Bradley–Terry	✗	asymptotic only	pairwise-only	✗	✗	✗
Factor analysis (1-factor)	loading ≈ discrimination	Gaussian posterior	EM handles MAR	partial	✗	✗
Classical IRT (flat)	✓	✓ (Bayesian)	MAR by construction	via equating	✗	✗
Multidimensional IRT	✓	✓	MAR	via equating	✗	unstructured factor vector
AICO	✓ (per benchmark, 1/2/3PL)	✓ (full posterior)	MAR by construction	anchor-based equating engine	✓ (5 likelihood families)	✓ (global→source→domain→b

Tradeoffs. AICO's advantages are bought with computation (a NUTS fit versus a mean), prior specification (defaults are weakly informative but exist), and engineering surface area. When the portfolio is small, dense, and static, the flat average is nearly as good at ranking (Section 8) — the gap opens exactly where enterprises live: sparse observation, heterogeneous benchmarks, churning portfolios, and decisions that need uncertainty.

4. Background and Mathematical Foundations

Full derivations, assumptions, and proofs are in Appendix A; this section states the model. Notation: models m = 1..M, sources s ∈ {public, internal}, domains d = 1..D (each nested in one source), benchmarks b = 1..B (each in one domain), metrics k = 1..K_b per benchmark, observed set $\mathcal{O}$ of (m, b, k) triples.

4.1 Hierarchical capability cascade

$$\theta_m \sim \mathcal{N}(0,1) \qquad \text{(global capability; the identification anchor)}$$

$$\gamma_{m,s} = \lambda_s\,\theta_m + v_{m,s}, \qquad v_{m,\cdot} \sim \mathcal{N}(0,\Sigma_s) \quad \text{(source)}$$

$$\delta_{m,d} = \lambda_d\,\gamma_{m,s(d)} + w_{m,d}, \qquad w_{m,\cdot} \sim \mathcal{N}(0,\Sigma_d) \quad \text{(domain)}$$

$$\beta_{m,b} = a_b(\delta_{m,d(b)} - d_b) + u_{m,b}, \qquad u_{m,b} \sim \mathcal{N}(0,\sigma_b^2)\cdot \mathbb{1}[K_b > 1] \quad \text{(benchmark, IRT edge)}$$

Residuals within the source and domain layers are correlated across coordinates via LKJ-Cholesky priors (public and internal capability co-vary beyond what θ explains). Loadings λ_s, λ_d and discriminations a_b carry LogNormal priors (positivity = reflection identification); difficulties d_b are partially pooled within domains; c_b ~ Beta(2, 18) is the 3PL guessing floor. irt_model ∈ {1PL, 2PL, 3PL} per benchmark selects the edge form (1PL pins a_b ≡ 1).

4.2 Observation layer (GLLTM)

Each metric k on benchmark b observes β_{m,b} through one of five links:

Family	Likelihood	Link
normal	y ~ N(η, σ_k²)	η = μ_k + λ_k β_{m,b}
beta	y ~ Beta(μφ_k, (1−μ)φ_k)	μ = logit⁻¹(μ_k + λ_k β_{m,b})
binomial	y ~ Bin(n, π)	π = c_b + (1−c_b)\,σ(β_{m,b})
bernoulli	y ~ Bern(π)	same as binomial
ordered-logistic	P(y ≤ c) = σ(κ_{k,c} − β_{m,b})	ordered cutpoints κ

All five are generalized linear models in the same scalar latent — a generalized linear latent-trait model — so mixed metric types coexist in one likelihood product and a missing observation is simply an absent factor (missing-at-random assumption; Appendix A.6).

4.3 Identifiability

The likelihood admits a one-parameter scale-and-reflection invariance (θ' = αθ, λ' = λ/α, a' = a/α, d' = αd). Four constraints remove it: (1) the fixed N(0,1) anchor prior on θ (location/scale), (2) positivity of all loadings and discriminations (reflection), (3) difficulty defined relative to the anchored domain scale, (4) the single-indicator constraint fixing the metric loading when K_b = 1. Appendix A.3 gives the proof; the implementation enforces each rule structurally and the test suite gates on an identifiability checklist.

4.4 Outputs

One posterior fit yields, per draw: standardized global capability $\theta^{std} = (\theta - \bar{\theta})/\mathrm{sd}(\theta)$, per-source and per-domain capabilities, per-benchmark capabilities, the generalization gap $\gamma_{m,public} - \gamma_{m,internal}$, benchmark difficulty/discrimination/guessing estimates, and induced rank distributions. Point summaries use posterior means with HDIs; all downstream engines consume draws, not points.

5. Architecture

AICO is a layered system; every layer is a typed Python package with its own tests.

```
observations ──► normalization ──► hierarchical Bayesian core (PyMC: NUTS / ADVI)
                     │ posterior draws
   ┌──────────────────────┬────────────────────────┼──────────────────────┬──────────────────────┐
   ▼            ▼          ▼          ▼          ▼
 scale stability  forecasting  attribution  portfolio opt.  capability graph
 (anchors,      (LLT Kalman, (Shapley,    (benchmark      (embeddings,
  equating, drift) GP, HMM)    decomposition) selection)      coverage gaps)
   └──────────────────────┴────────────────────────┴──────────────────────┴──────────────────────┘
                     ▼
        executive decision intelligence (MCDA, scenarios, procurement)
                     ▼
       reports / API server / MCP assistant / observability
```

5.1 Bayesian capability core (bayesian/, irt/, api/): builds the Section 4 model from validated records, fits via NUTS (default: 4 chains) or ADVI, checks convergence ($\hat{R}$, ESS, divergences; ELBO trace for VI), and extracts every output.

5.2 Capability scale stability (stability/, onboarding/): anchor-model designs and equating transforms keep the capability scale comparable across portfolio changes; drift detectors flag when re-equating is needed. New benchmarks are onboarded by calibrating their IRT parameters against the existing scale before they influence it.

5.3 Capability forecasting (forecasting/): local-linear-trend Kalman filters per model/domain with closed-form Gaussian forecasts; GP forecasts for nonlinear trajectories; Gaussian HMMs and structural-break tests for regime changes; saturation risk from frontier-curve fits (logistic/power-law/exponential, AIC-selected); threshold- crossing probability estimates.

5.4 Capability attribution (attribution/): decomposes capability changes between fits into benchmark- and domain-level contributions; Shapley attribution over benchmark coalitions provides an efficiency-satisfying decomposition (attributions sum to the total change).

5.5 Benchmark portfolio optimization (optimization/): selects benchmark subsets maximizing information about the capability layers under evaluation-cost budgets — identifying redundant benchmarks and coverage gaps jointly with 5.6.

5.6 Capability intelligence graph (graph/): a typed graph of models, benchmarks, domains, and capabilities with GNN embeddings; supports similarity queries, coverage-gap detection, and structural analytics.

5.7 Executive decision intelligence (decision/, reports/): multi-criteria decision analysis over posterior capability distributions (procurement, deployment gates), scenario analysis over forecast distributions, and executive report generation.

5.8 Platform (server/, observatory/, integrations/, observability/, governance/): FastAPI service, storage plugins (in-memory, Postgres), MCP server and OpenWebUI tool integrations, OpenTelemetry/Prometheus observability, and governance (registries, audit trails, reproducibility records).

6. Simulation Studies

Design. The simulation framework (simulation/) generates synthetic ecosystems by forward-sampling the fitted model's own prior (pm.sample_prior_predictive on the real model graph), retaining every latent draw as ground truth. This eliminates generator/model mismatch as a confound in recovery studies: the data are exactly one draw from the assumed data-generating process, so failures are attributable to inference, not misspecification. (Robustness to misspecification is tested separately via prior sensitivity analysis and PPCs on real data; Section 10 discusses the limitation.)

Scales are named presets: aico-100 (100 models × 100 benchmarks, 60% observed), aico-1k ($10^3 \times 10^3$, 5% observed), aico-10k ($10^4 \times 10^4$, 0.2% observed) — density falls with scale, matching real ecosystems — plus ci-small (12 × 18) for continuous validation. Domain counts, metric-type mixtures (all five families), IRT-model mixtures, multi-metric fractions, and difficulty distributions are all parameterized (EcosystemConfig).

Measurements per replication (run_simulation_study): recovery of θ (correlation, RMSE, bias against truth), ranking accuracy (Spearman/Kendall vs. truth), uncertainty calibration (empirical central-interval coverage at 50/90/95% — nominal by construction under correct inference), scale stability (rank agreement between the full fit and a refit on a random 80% benchmark subset), and forecast quality on simulated capability trajectories anchored at the true capabilities (hold-out MAE, RMSE, 95% coverage, Gaussian CRPS). Aggregates are mean ± sd over independent replications (re-seeded ecosystems, not re-fits of one dataset).

Representative results (ci-small, ADVI, seeds 0–1; regenerate with scripts/run_scientific_validation.py): recovery correlation 0.74 ± 0.08, ranking Spearman 0.80 ± 0.10, 95% coverage 0.92 ± 0.00, scale stability 0.99 ± 0.00, forecast 95% coverage 1.00. Two findings are worth flagging as expected and diagnostic: ADVI under-covers at the 50% level (0.33 vs. nominal 0.50) — the well-known variational variance underestimation — while NUTS runs are calibrated at all levels; and recovery correlation at 12 models is bounded by the small-M posterior shrinkage toward the anchor prior, rising with scale. The committed CI baseline (scripts/scientific_baseline.json) pins these numbers; drift beyond ±0.15 fails the build.

7. Scientific Validation

The validation framework (validation/) implements, and CI continuously runs:

- Parameter recovery (recovery): posterior vs. truth for θ (global, domain), discrimination, difficulty; correlation/RMSE/bias/95%-coverage per family.
- Capability recovery & rank stability (rank_stability): posterior rank distributions; stability under data perturbation.
- Posterior calibration (calibration): PIT uniformity tests.
- Coverage probability (coverage): repeated simulate-fit-check of HDI coverage with binomial hypothesis tests, tolerant of isolated numerical failures.
- Posterior predictive checks (ppc): per-metric-channel test statistics and reliability diagrams.

- Model comparison (comparison): WAIC and PSIS-LOO over the concatenated pointwise log-likelihood across heterogeneous metric channels.
- Prior sensitivity (sensitivity): refits under perturbed prior scales.
- Benchmark drift detection (stability/drift): monitors equated scales.
- Generalization-gap stability: the public-vs-internal gap is a first-class output tracked across refits.
- External validation (validation/external; Section 9.3).

The continuous scientific validation gate (scripts/run_scientific_validation.py, .github/workflows/scientific-validation. runs a full simulation study nightly and on any PR touching model/inference code, enforcing hard floors (recovery correlation, ranking Spearman, coverage, scale stability, forecast quality) and drift tolerance against the committed baseline. A model or dependency change that silently degrades scientific quality fails CI.

8. Ablation Studies

The ablation framework (experiments/ablation.py) removes one component at a time, replacing it with the naive alternative a team would otherwise use, and measures degradation on ground-truth ecosystems (mean over replications; regenerate via run_ablation_study):

Component removed	Replacement	Degradation metric(s)
IRT edge	discrimination ≡ 1, difficulty ≡ 0 (pinned priors)	ranking Spearman ↓, recovery RMSE ↑
Hierarchy / latent model	flat per-benchmark z-score averaging	ranking Spearman ↓, top-k regret ↑, posterior lost entirely
Forecasting	persistence forecast	hold-out MAE +89%, 95%-coverage error +0.41 (illustrative ci-small run)
Stable latent scale	raw-average ranking	rank stability under 20% benchmark retirement ↓
Decision intelligence	select top-k by flat average	selection regret ↑

Degradations grow with ecosystem sparsity and heterogeneity (the IRT and hierarchy ablations are nearly free on dense, homogeneous data — consistent with Section 3.6's tradeoff summary). Graph intelligence and the executive layer are consumers of posterior outputs; their "ablation" removes functionality (similarity queries, coverage-gap detection, risk-aware recommendations) rather than degrading a shared scalar metric, and they are therefore evaluated by the decision-regret proxy and qualitatively.

9. Benchmark Comparisons

9.1 Harness. baselines/ implements mean, weighted mean, median, Elo, TrueSkill (two-player Gaussian updates), Bradley–Terry (MM algorithm), and single-factor analysis (EM over observed cells) on a common standardized score-matrix interface, plus a PrecomputedScores adapter that lets AICO's posterior means enter the same harness. The harness measures: Spearman/Kendall vs. truth; RMSE after affine alignment; empirical 95% coverage for estimators that report uncertainty; ranking accuracy at 100/50/25/10% observation density; self-consistency under 20% observation dropout; and rank stability under benchmark retirement/addition.

9.2 Findings (synthetic one-factor and hierarchical ecosystems; regenerate with run_baseline_comparison): (i) on dense data all methods rank well (Spearman ≥ 0.97 on the illustrative 30 × 40 run) — averaging is a strong baseline when its assumptions hold; (ii) pairwise methods (Elo, TrueSkill, Bradley–Terry) degrade fastest under sparsity (Spearman 0.65–0.70 at 10% density vs. 0.92 for the mean family), because reducing scores to comparisons discards magnitude exactly when data are scarce; (iii) factor analysis matches AICO's point-ranking on flat data but has no difficulty parameter, no hierarchy, and degrades sharply at extreme sparsity (0.45 at 10%); (iv) only TrueSkill and FA report uncertainty at all, and both under-cover (≈0.90 at nominal 0.95 on the illustrative run) without the hierarchical pooling that keeps AICO's intervals calibrated; (v) under benchmark retirement, latent-variable methods (FA, AICO) hold rank stability best. AICO's advantage is thus conditional and quantified: uncertainty, difficulty-awareness, and portfolio-change stability, at higher compute cost.

9.3 External validation. validation/external implements the replication protocol for real suites: (a) reference-ranking agreement — Spearman/Kendall/top-k overlap against an independently published ordering, where moderate disagreement is expected (the reference is usually itself a flat average) and large unexplained disagreement is investigated; (b) held-out benchmark prediction — fit on a benchmark subset, test whether estimated capabilities rank models correctly on each held-out benchmark (the external analogue of PPC, and the sharpest test that a latent capability rather than benchmark memorization is measured). The methodology, including manifest requirements for independent replication, is documented in docs/science/external-validation.md.

10. Threats to Validity and Limitations

Internal. Simulation studies draw data from the model's own prior — the best case for recovery. Prior-sensitivity analysis and PPCs mitigate but do not eliminate misspecification risk; external validation (9.3) is the real check. VI results systematically under-cover at inner quantiles — mean-field ADVI's factorized approximation underestimates posterior variance, and the effect is measurable in our own CI baseline (scripts/scientific_baseline.json: 50% HDI empirical coverage ≈ 0.33 and 95% ≈ 0.88 on the ci-small preset under ADVI). VI-based intervals must therefore be treated as anticonservative; we report VI as the fast path, gate calibration claims on NUTS, and every API response carrying interval estimates names the inference method that produced it so downstream consumers can apply the same discipline.

Construct. "Capability" is defined by the benchmark portfolio; a biased portfolio yields a precisely-measured biased construct. The portfolio-optimization and coverage-gap layers surface, but cannot resolve, portfolio bias. Benchmark contamination appears as inflated β on affected benchmarks; drift detection flags scale anomalies but cannot prove contamination.

Statistical. MAR is assumed; evaluation choices correlated with expected performance (publication bias) violate it. The static model treats observations as exchangeable in time within a fit; temporal structure is handled by refits plus the forecasting layer rather than the (specified, unimplemented-by-default) temporal extension.

Scale. NUTS at aico-10k is expensive; the practical path at that scale is ADVI plus NUTS on anchored subsets. Performance benchmarks (benchmarks/) quantify the frontier.

11. Future Work

The long-term agenda is maintained in docs/roadmap.md: adaptive/computerized-adaptive benchmark selection; native multidimensional capability vectors with structured priors; first-class temporal hierarchical modeling; cross-organizational and federated capability estimation with privacy budgets; causal capability inference (training-intervention effects); active benchmark generation and synthesis; evaluator (LLM-judge) reliability modeling as a measurement layer; richer forecast uncertainty (deep state-space models); and benchmark-governance economics.

12. Conclusion

Capability measurement is a measurement problem, and measurement problems have a century of statistical machinery. AICO applies that machinery — hierarchical Bayes, IRT, equating, state-space forecasting, proper scoring rules — to enterprise AI evaluation, wraps it in production engineering, and holds itself to the standard it proposes: every number reproducible, every claim gated in CI, every component ablated and compared against the baselines it claims to beat.

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Appendix A — Mathematical Appendix

A.1 Notation

Symbol	Meaning
M, S, D, B, K_b	counts of models, sources, domains, benchmarks, metrics on benchmark b
θ_m	global capability of model m (anchor scale)
γ_{m,s}, δ_{m,d}, β_{m,b}	source / domain / benchmark capability

Symbol	Meaning
λ_s, λ_d	source / domain loadings (> 0)
a_b, d_b, c_b	benchmark discrimination (> 0), difficulty, guessing floor
Σ_s, Σ_d	LKJ-Cholesky residual covariances across sources / co-parental domains
σ_b	testlet residual sd (only $K_b > 1$)
μ_k, λ_k, σ_k, φ_k, $\kappa_{k,\cdot}$	metric intercept, loading, residual sd, beta precision, ordered cutpoints
$\mathcal{O}$	observed (m,b,k) index set; $y_{\mathcal{O}}$ observed values
s(d), d(b)	parent maps (domain→source, benchmark→domain)
$\sigma(\cdot)$, $\Phi(\cdot)$, $\varphi(\cdot)$	logistic function, standard normal cdf, pdf

A.2 Joint density, likelihood, and priors

The joint posterior over all latents Θ and hyperparameters Λ:

$$p(\Theta, \Lambda \mid y_{\mathcal{O}}) \propto \prod_m \mathcal{N}(\theta_m; 0, 1) \prod_m \mathcal{N}(v_{m,\cdot}; 0, \Sigma_s) \prod_m \mathcal{N}(w_{m,\cdot}; 0, \Sigma_d) \prod_{m,b:K_b>1} \mathcal{N}(u_{m,b}; 0, \sigma_b^2) \times \prod_{(m,b,k)\in\mathcal{O}} F_{\tau(}$$

with deterministic cascade $\gamma = \lambda_s \theta + v$, $\delta = \lambda_d \gamma_{s(d)} + w$, $\beta = a_b(\delta_{d(b)} - d_b) + u$, observation families F from Section 4.2, and priors

$$\lambda_s, \lambda_d \sim \text{LogNormal}(0, 0.5), \quad a_b \sim \text{LogNormal}(0, 0.5), \quad d_b \sim \mathcal{N}(\bar{d}_{d(b)}, \tau^2_{d(b)}),\ \bar{d}_d \sim \mathcal{N}(0, 1.5),\ \tau_d \sim \text{HalfNorm}$$

$$c_b \sim \text{Beta}(2, 18), \quad \sigma_b \sim \text{HalfNormal}(0.25), \quad \Sigma_\cdot = \text{diag}(\sigma)\, C\, \text{diag}(\sigma),\ C \sim \text{LKJ}(\eta{=}2),\ \sigma \sim \text{HalfNormal}(0.3),$$

$$\mu_k \sim \mathcal{N}(0, 1), \quad \lambda_k \sim \text{LogNormal}(0, 0.3)\,(K_b > 1), \quad \sigma_k \sim \text{HalfNormal}(0.5), \quad \phi_k \sim \text{Gamma}(2, 0.1),$$

with ordered-logistic cutpoints given increment-based ordered priors (increment sd 1.5). All scales are configuration (PriorConfig), not constants.

Assumptions. (i) Conditional independence of observations given β; (ii) MAR missingness (A.6); (iii) linearity of the source/domain edges with Gaussian residuals; (iv) a single dominant capability dimension per layer coordinate (multidimensionality is expressed hierarchically, not as a free factor vector); (v) exchangeability of models under the anchor prior.

A.3 Identifiability

Proposition. Under Rules 1–4 below, the map $\Theta \mapsto L(\Theta \mid y)$ has no non-trivial invariance, and the posterior is proper.

The invariance. For $\alpha \neq 0$ define T_α: $\theta' = \alpha\theta$, $\lambda_s' = \lambda_s/\alpha$, $\lambda_d' = \lambda_d/\alpha$, $a_b' = a_b/\alpha$, $d_b' = \alpha\, d_b$ (residuals scaled accordingly). Substitution shows every linear predictor is invariant: $\lambda_s'\theta' = \lambda_s\theta$, and since δ inherits the factor α through the zero-intercept cascade, $a_b'(\delta' - d_b') = a_b(\delta - d_b)$. Every likelihood family depends on latents only through β (or through μ_k + λ_kβ with free μ_k, λ_k), hence L(T_α(Θ) | y) = L(Θ | y). Because the upper edges carry no intercepts, the invariance group is exactly this one-parameter scale-and-reflection family (not the full affine group of unrestricted factor models).

Resolution. Rule 1 (anchor): θ ~ N(0,1) with fixed hyperparameters. The prior is not T_α-invariant for α ≠ 1, so the posterior breaks scale invariance (Bafumi et al., 2005). Rule 2 (reflection): LogNormal priors restrict λ_s, λ_d, a_b > 0, excluding α < 0. Combined with Rule 1, the residual group is trivial. Rule 3 (difficulty): with the domain scale fixed transitively by Rules 1–2, d_b is identified as the domain-scale location where the IRT edge crosses zero — a parameter, not a competing anchor. Rule 4 (single indicator): when K_b = 1, a_b and λ_k would multiply against one data column (classical SEM single-indicator indeterminacy); fixing λ_k = 1, μ_k = 0 (and σ_b = 0) removes it. ∎

Practical identifiability (distinct from structural): with few observations per model, the posterior concentrates slowly along near-flat directions (e.g. a_b vs. σ_b on multi-metric benchmarks). The implementation gates on an identifiability checklist and convergence diagnostics; reporting uses the per-draw standardization θ^std = (θ − θ̄)/sd(θ), which is invariant to any residual prior-induced drift.

A.4 Inference and convergence

NUTS (Hoffman & Gelman, 2014) with non-centered parameterizations for all Gaussian latents; convergence assessed by split-$\hat{R}$ < 1.01, bulk/tail ESS thresholds, and zero post-warmup divergences (Vehtari et al., 2021 recommendations). ADVI (Kucukelbir et al., 2017) is the fast path: a mean-field Gaussian approximation fit by stochastic ELBO maximization; its known variance underestimation is measured in the simulation studies (inner-quantile under-coverage) rather than assumed away. Geometric ergodicity of NUTS on this model class is not proven (typical for hierarchical GLMMs); we rely on the standard empirical convergence battery, which is exactly what the continuous validation gate automates.

A.5 Score definitions

Per posterior draw t:

$$\theta_m^{\mathrm{std},(t)} = \frac{\theta_m^{(t)} - \overline{\theta^{(t)}}}{\mathrm{sd}(\theta^{(t)})}, \qquad \mathrm{ICI}_{m,\ell}^{\mathrm{layer}} = \text{posterior functional of } (\gamma, \delta, \beta),$$

$$\mathrm{GenGap}_m^{(t)} = \gamma_{m,\mathrm{public}}^{(t)} - \gamma_{m,\mathrm{internal}}^{(t)}, \qquad \mathrm{rank}_m^{(t)} = 1 + \#\{m' : \theta_{m'}^{(t)} > \theta_m^{(t)}\}.$$

Point estimates are posterior means; intervals are HDIs; rank distributions are reported whole (rank point estimates alone are famously misleading; cf. posterior rank stability in validation/rank_stability).

A.6 Missing data

Let R be the observation indicator over model-benchmark-metric triples. AICO assumes MAR: p(R | y, X) = p(R | y_$\mathcal{O}$, X). Under MAR the observed-data likelihood ∏_{(m,b,k)∈$\mathcal{O}$} F(y | β) is valid for Bayesian inference without modeling R (Rubin, 1976). Violations (evaluations chosen because a model is expected to do well) bias layer estimates toward the observed regime; the coverage-gap analytics in the graph layer surface systematically unobserved regions, and the sensitivity analysis quantifies prior dependence in sparse regions.

A.7 Scale stability and equating

Let fits F_1, F_2 share anchor models $\mathcal{A}$ (evaluated in both). The equating transform is the affine map g(x) = ax + c minimizing ∑_{m∈$\mathcal{A}$} (g(θ_m^{F_2}) − θ_m^{F_1})^2 (mean–sigma equating; Kolen & Brennan, 2014), applied to all F_2 scores; drift statistics test whether ‖(a, c) − (1, 0)‖ exceeds sampling noise (bootstrap over anchors). Anchor sets are chosen to span the capability range (onboarding/anchors); new benchmarks are calibrated with capabilities frozen, then released into the joint fit.

A.8 Forecasting

Local linear trend per series: state x_t = (level, trend),

$$x_{t+\Delta} = \begin{pmatrix} 1 & \Delta \\ 0 & 1 \end{pmatrix} x_t + w_t, \quad w_t \sim \mathcal{N}(0, Q(\Delta))\,, \qquad y_t = (1\;0)\,x_t + \varepsilon_t,$$

with the exact discretization of the continuous-time process (white level noise of intensity σ_ℓ^2 plus an integrated-Wiener trend of intensity σ_τ^2):

$$Q(\Delta) = \begin{pmatrix} \sigma_\ell^2\Delta + \sigma_\tau^2\Delta^3/3 & \sigma_\tau^2\Delta^2/2 \\ \sigma_\tau^2\Delta^2/2 & \sigma_\tau^2\Delta \end{pmatrix}.$$

The off-diagonal and cubic terms are what keep long-horizon level uncertainty honest — a diagonal $\mathrm{diag}(\Delta\sigma_\ell^2, \Delta\sigma_\tau^2)$ approximation omits the trend noise that integrates into the level and understates forecast variance as Δ grows. Exactness also makes one step of size Δ identical to any subdivision of it. Kalman-filtered; h-step forecasts are the closed-form Gaussian push-forward of the final state (latent level, observation noise excluded). GP forecasts use squared-exponential plus linear kernels with marginal-likelihood hyperparameters; regime changes use Gaussian HMMs (forward-backward + Viterbi) and CUSUM-style structural-break tests; saturation risk comes from AIC-selected logistic/power-law/exponential frontier fits. Forecast quality is scored by MAE, RMSE, central-interval coverage, and Gaussian CRPS CRPS(N(μ,σ^2), y) = σ[z(2Φ(z)−1) + 2φ(z) − 1/$\sqrt{\pi}$], z = (y−μ)/σ (Gneiting & Raftery, 2007).

A.9 Attribution

For a capability change Δ = f(P_2) − f(P_1) between portfolio states, benchmark b's Shapley attribution is

$$\phi_b = \sum_{S\subseteq\mathcal{B}\setminus\{b\}} \frac{|S|!\,(|\mathcal{B}| - |S| - 1)!}{|\mathcal{B}|!}\,[f(S\cup\{b\}) - f(S)],$$

estimated by permutation sampling with convergence monitoring; efficiency (∑_b φ_b = Δ) holds by construction and is the property naive marginal attribution violates (measured in the ablation study). Coarser exact decompositions by domain/source are available where the layer structure makes f additive.

A.10 Portfolio optimization

Benchmark selection maximizes expected information about the capability layers under a budget: max_{S ⊆ $\boldsymbol{\mathcal{B}}$, cost(S) ≤ C} I(S), with I an information proxy (expected posterior variance reduction for target layers, estimated from the fitted model's discrimination/noise parameters). The objective is monotone submodular under the Gaussian approximation, so greedy selection carries the (1 − 1/e) guarantee (Nemhauser et al., 1978); the implementation supports greedy and exact small-portfolio solvers plus redundancy scoring.

A.11 Decision functions

Procurement/deployment decisions are expected-utility rules over the posterior: for action a with utility $U(a, \theta)$, $a^* = \arg\max_a E_{p(\theta|y)}[U(a, \theta)]$, with MCDA weights eliciting U and scenario analysis replacing $p(\theta|y)$ by forecast push-forwards at horizon h. Threshold questions use $P(\theta_m \geq \tau \mid y)$ directly from draws. The paper's evaluation metric is selection regret: regret(k) = mean-true-capability(oracle top-k) − mean-true-capability(selected top-k).

A.12 Additional references (appendix)

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Appendix B — Experimental Protocol

Every table in Sections 6–9 is produced by experiments.run_experiment, which writes the report next to a JSON manifest recording: git commit (and dirty flag), Python version, versions of every numerically relevant package (numpy, scipy, pymc, pytensor, arviz, pandas, xarray, networkx), OS/kernel/architecture/CPU count, container image (when run in the shipped Docker image, exported as AICO_CONTAINER_IMAGE), all seeds, and content hashes of input datasets. Commands:

```
# Simulation studies (Section 6) + CI gates (Section 7)
python scripts/run_scientific_validation.py --preset aico-100 --method nuts --seeds 0 1 2

# Ablations (Section 8): experiments.run_ablation_study via a driver of your choice
# Baseline comparisons (Section 9): baselines.run_baseline_comparison
# Performance benchmarks (Section 10 discussion): benchmarks/run_performance_benchmarks.py
# Open datasets used in notebooks/tutorials: scripts/generate_open_datasets.py
```

Independent replication requires only the repository at the manifest's commit, Python 3.13, and the pinned dependency versions; datasets regenerate byte-identically from seeds (hash-checked in tests/test_generate_open_datasets_script.py).